Transplantation and Cellular Therapy 31 (2025) 668.e1−668.e12

Transplantation and Cellular Therapy

journal homepage: www.astctjournal.org

American Society for Transplantation and Cellular Therapy

Full Length Article
Biomarkers

Inflammatory Markers and Microbiome Dysbiosis in Hematopoietic Cell Transplant Recipients with Lung Graft-versus-Host Disease

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Article history:
Received 11 March 2025
Accepted 12 June 2025

Key Words:
Biomarkers
Bronchiolitis obliterates
Chronic graft-versus-host disease
Gut microbiome

A B S T R A C T

Bronchiolitis obliterans (BOS) is a manifestation of pulmonary chronic graft-versus-host disease (cGVHD) and is a devastating complication of allogeneic hematopoietic stem cell transplantation (HCT). Early detection and treatment of BOS may improve outcomes, but biomarkers that accurately identify BOS early are lacking. We aimed to determine whether certain validated cGVHD markers could also accurately diagnose BOS as compared with patients without BOS and with or without extrapulmonary cGVHD. In addition, we sought to determine whether dysbiosis of the gut or oral microbiomes was associated with BOS or with inflammatory biomarkers. We enrolled 43 recipients of allogeneic HCT, 16 of whom

Financial disclosure: See Acknowledgments on page 668.e11.

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https://doi.org/10.1016/j.jtct.2025.06.020
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Hematopoietic stem cell transplant

had BOS. For each patient, we obtained pulmonary function tests, measured the levels of 9 serum biomarkers utilizing enzyme-linked immunosorbent assays, and analyzed both the oral and gut microbiome using microbial DNA amplification and sequencing. We compared biomarker levels to lung function, both at baseline and over time, as well as to microbiome diversity. Higher IL1RL1 (P = .002) and IL-17 (P = .041) at enrollment were negatively correlated with FEV1% (forced expiratory volume in 1 second) lung function over time. Increases in IL1RL1 (P = .035), IL-17 (P = .009), and WFDC2 (P = .045) levels over time were associated with worsened lung function/FEV1% over time. There were minimal correlations between gut microbiome diversity and lung function or serum biomarkers. Oral microbiome alpha diversity was lower in subjects with BOS than without (P = .00057), and oral beta diversity was associated with FEV1% and with levels of several biomarkers. Our pilot study suggests that certain serum cGVHD markers may identify recipients of allogeneic HCT at higher risk for pulmonary impairment over time and that these markers should be followed with robust, controlled studies.

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INTRODUCTION

Although allogeneic hematopoietic cell transplantation (HCT) has significantly altered the course of treatment for many hematologic malignancies, alloimmune injury to recipient organs in the form of chronic graft-versus-host disease (cGVHD) remains a devastating complication [1]. Bronchiolitis obliterans syndrome (BOS) is a severe pulmonary manifestation of cGVHD that results in progressive airflow obstruction and is associated with substantial morbidity and mortality [2−4]. Treatment options are limited, but outcomes may be improved with early detection and initiation of immunosuppression [5,6].

Pulmonary function tests (PFTs) are routinely used at our institutions after HCT to screen for early BOS [7,8]. However, a limitation of this approach is the high false-positive rate, particularly if relying on a single abnormal PFT [7,9]. While biomarkers that accurately identify BOS, particularly early in its course, might ameliorate this issue, such biomarkers are currently lacking. The identification of noninvasive biomarkers to distinguish BOS, particularly early BOS, from other forms of pulmonary impairment would significantly improve the capability of surveillance PFTs and home spirometry to identify patients with BOS before significant airflow obstruction has occurred.

While several serum biomarkers have been identified as markers of cGVHD, none have been specifically identified as markers of BOS that might distinguish BOS from systemic inflammation due to extensive cGVHD. Some biomarkers, such as MMP3 and MMP9, may be elevated in patients with post-HCT BOS [10,11]. However, few studies have focused on newly diagnosed BOS [11], which may have a different immunologic profile than longstanding BOS, and none have examined how these biomarkers influence pulmonary impairment over time. The primary goal of this work was to serve as a prospective pilot study to determine whether certain validated cGVHD markers were more likely to be found in patients with cGVHD with BOS as compared with patients with cGVHD but without BOS. Secondary objectives were to determine whether serum markers could identify early BOS and whether serum markers were associated with lung function decline 1 year after the onset of impairment. An exploratory objective was to determine whether the dysbiosis of the gut or oral microbiome was associated with BOS or inflammatory markers. Ultimately, we aimed to identify candidate biomarkers that could be validated in ongoing prospective studies of BOS [12,13], paving the way for diagnostic biomarkers that facilitate early diagnosis.

MATERIALS AND METHODS

Study Participants

We prospectively enrolled 43 recipients of allogeneic HCT at MD Anderson Cancer Center (Houston, TX) between 2018 and 2021. Participants were included if they were 18 years of age or older at enrollment, received their transplant within 5 years of the enrollment date, and had a PFT prior to study enrollment. Subjects were excluded if they had evidence of acute lower respiratory tract infection, active respiratory viral infection, or a past history of obstructive or restrictive lung disease. Our study exclusion criteria were intentionally minimal in order to maximize the number of patients with BOS that we would be able to identify early in their disease course. Relapse was not considered to be an exclusion, but patients who developed a relapse were not monitored further and were censored. Data on prior antibiotic exposure, respiratory infection, and atopy were obtained via chart review, while data pertaining to HCT regimen and cGVHD were obtained from an

Table 1
Demographics of Study Participants

	Group				
	Total (N = 43)	Control (n = 12)	Pre-BOS (n = 6)	Transient pulmonary impairment (n = 9)	BOS (n = 16)
Female sex	24 (55.8%)	5 (41.7%)	2 (33.3%)	3 (33.3%)	14 (87.5%)
Race					
White	28 (65.1%)	10 (83.3%)	5 (83.3%)	4 (44.4%)	9 (56.2%)
Black	1 (2.3%)	0	0	0	1 (6.2%)
Hispanic	8 (18.6%)	2 (16.7%)	0	1 (11.1%)	5 (31.2%)
Other	6 (13.9%)	0	1 (16.7%)	4 (44.4%)	1 (6.2%)
Underlying Malignancy					
Acute myeloid leukemia	10 (23.2%)	1 (8.3%)	2 (33.3%)	2 (22.2%)	5 (31.2%)
Acute lymphoblastic leukemia	9 (20.9%)	1 (8.3%)	0	3 (33.3%)	5 (31.2%)
Chronic myeloid leukemia	3 (7%)	0	1 (16.7%)	1 (11.1%)	1 (6.2%)
Chronic lymphocytic leukemia	3 (7%)	2 (16.7%)	0	0	1 (6.2%)
Myelodysplastic syndrome	8 (18.6%)	3 (25%)	2 (33.3%)	1 (11.1%)	2 (12.5%)
Myelofibrosis	3 (7%)	3 (25%)	0	0	0
Aplastic anemia	1 (2.3%)	1 (8.3%)	0	0	0
Non-Hodgkin lymphoma	1 (2.3%)	0	1 (16.7%)	0	0
Angioimmunoblastic T cell lymphoma	1 (2.3%)	0	0	1 (11.1%)	0
Multiple myeloma	1 (2.3%)	0	0	0	1 (6.2%)
Unknown	3 (7%)	1 (8.3%)	0	1 (11.1%)	1 (6.2%)
Age at transplant (median, interquartile range)	53, 27.5	60.5, 9	47.5, 24.75	38, 25	49.5, 23.25
Cell Source					
Peripheral	31 (72.1%)	10 (83.3%)	5 (83.3%)	4 (44.4%)	12 (75%)
Bone marrow	11 (25.6%)	2 (16.7%)	1 (16.7%)	5 (55.5%)	3 (18.7%)
Donor Type					
Matched related	18 (41.9%)	5 (41.7%)	2 (33.3%)	4 (44.4%)	7 (43.7%)
Matched unrelated	20 (46.5%)	5 (41.7%)	4 (66.7%)	5 (55.5%)	6 (37.5%)
Haploidentical	4 (9.3%)	2 (16.7%)	0	0	2 (12.5%)
Donor Sex–Recipient Sex Matching					
Female to Male	8 (18.6%)	4 (33.3%)	1 (16.7%)	3 (33.3%)	0
Male to Female	10 (23.2%)	2 (16.7%)	0	1 (11.1%)	7 (43.7%)
Sex matched	22 (51.2%)	5 (41.7%)	5 (83.3%)	4 (44.4%)	8 (50%)
Unknown	3 (7%)	1 (8.3%)	0	1 (11.1%)	1 (6.2%)
Myeloablative therapy used—yes	31 (72%)	5 (41.7%)	5 (83.3%)	8 (88.9%)	13 (81.3%)
Extrapulmonary cGVHD at time of enrollment					
Yes	24 (55.8%)	6 (50%)	5 (83.3%)	1 (11.1%)	12 (75%)
No	18 (41.9%)	6 (50%)	1 (16.7%)	7 (77.8%)	0
Unknown	1 (2.3%)	0	0	1 (11.1%)	
Site of extrapulmonary cGVHD					
Ocular	14 (32.6%)	4 (33.3%)	1 (16.7%)	0	9 (56.3%)
Skin	18 (41.9%)	3 (25%)	5 (83.3%)	1 (11.1%)	9 (56.3%)
Oral	10 (23.3%)	3 (25%)	1 (16.7%)	0	6 (37.5%)
Fascial	4 (9.3%)	0	1 (16.7%)	0	3 (18.7%)
Genital	7 (16.3%)	2 (16.7%)	0	0	5 (31.3%)

(*continued*)

Table 1 (*Continued*)

	Group				
	Total (N = 43)	Control (n = 12)	Pre-BOS (n = 6)	Transient pulmonary impairment (n = 9)	BOS (n = 16)
Gut	6 (13.9%)	0	2 (33.3%)	0	4 (25%)
Liver	3 (7%)	1 (8.3%)	1 (16.7%)	0	1 (6.3%)
Joint	2 (4.6%)	0	0	5 (55.5%)	2 (12.5%)
Antibiotic exposure within 30 days of enrollment	25 (58.1%)	6 (50%)	5 (83.3%)		9 (56.2%)
Steroid exposure within 30 days of enrollment	25 (58.1%)	3 (25%)	6 (100%)	4 (44.4%)	12 (75%)
Months since HCT (median, interquartile range)	15, 21	14, 10	24.5, 38	15, 50.25	14, 14.5
FEV1% at time of enrollment (mean ± standard deviation)	77.5 ± 20.6	100 ± 11.99	79.25 ± 7.5	83.3 ± 7.2	57.4 ± 16.6
FVC% at time of enrollment (mean ± standard deviation)	82.5 ± 13.7	96.55 ± 13	83.5 ± 6.9	82.1 ± 9.4	72.6 ± 10.4
DLCO at time of enrollment enrollment (mean ± standard deviation)	81.2 ± 16	85.75 ± 13.2	87.6 ± 10.4	79.2 ± 15.4	76.8 ± 19.9

cGVHD indicates chronic graft-versus-host disease; DLCO, diffusing capacity for carbon monoxide; FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; HCT, hematopoietic cell transplantation.

institutional HCT database. Basic demographic data is found in Table 1 as are further details regarding cGVHD manifestations and baseline lung impairment. Study approval was obtained from the MD Anderson institutional review board (IRB 2018-0288). Data were collected and analyzed in compliance with the Health Insurance Portability and Accountability Act.

Pulmonary Function Assessment

Upon enrollment in the study, we determined the clinical diagnosis, time from transplant to diagnosis, and severity of lung disease for each subject. Subjects were categorized as having BOS, BOS0p, or no BOS (control) based upon PFT results obtained at the time of enrollment. Patients with BOS must have met all diagnostic criteria per the 2014 NIH BOS Consensus Group [8]. Patients in the BOS0p group were defined as those who had a forced expiratory volume in 1 second (FEV1) decline of ≥10% but did not otherwise meet BOS criteria. This group was further subdivided into those who had an irreversible decline in FEV1 (pre-BOS) and those who demonstrated a decline in FEV1 by 10% with a return to baseline values by month 6 (transient pulmonary impairment). Forced vital capacity (FVC) and diffusing capacity for carbon monoxide (DLCO) were also recorded for each subject. All subjects in the control group had no evidence of impairment of the lungs on enrollment and did not demonstrate declines in pulmonary function. Our study included 16 subjects in the BOS group, 6 subjects with pre-BOS, 9 subjects with transient pulmonary impairment, and 12 controls.

Sample Collection

Upon enrollment in the study and every 3 months thereafter, serum, stool, and oral saliva samples were obtained from participants. We collected 20 mL of blood from each patient and centrifuged to obtain serum. We used single-plex enzyme-linked immunosorbent assays (ELISAs) to analyze serum for levels of several biomarkers that were identified *a priori* owing to their known association with cGVHD or respiratory failure post-HCT (WFDC2, DKK3, IL1RL1 [previously ST2], CCL5, CXCL9, CD163, IL-17, CCL15, and MMP3). ELISAs were performed using commercially available kits and were blinded to clinical information and reference standard results. ELISA procedures and parameters for the markers used herein have been previously described, and biomarker details can be found in Supplementary Table 3. [14,15]. Stool was collected in a standardized stool catcher provided to each patient, stored in a 4-mL sterile tube by the subject, and returned to the study facility where it was stored in a 4° C refrigerator.

Microbiome Analysis Methods

Microbial DNA extracted from stool samples as well as from oral rinse samples was amplified by polymerase chain reaction and then sequenced on the Illumina MiSeq platform (San Diego, CA) using paired-end protocols. Additional amplification details are located in the Supplementary Methods section. After sequencing, the reads were merged, dereplicated, and length-filtered using VSEARCH. Following denoising and chimera calling using the UNOISE3 commands (https://www.biorxiv.org/content/10.1101/081257v1), the unique sequences, or zero-radius OTUs (ZOTUs), were taxonomically classified using mothur [16] with the SILVA database version 138. Alpha and beta diversity metrics were generated in QIIME 2.

Statistical Analyses

All data analyses were performed using R software (version 4.3.2). A similar analytical approach was applied to both stool and oral samples. Alpha diversity was computed using the inverse Simpson index and compared across groups using the Wilcoxon or Kruskal-Wallis test. Beta diversity was computed using weighted normalized Unifrac and compared across groups using PERMANOVA [17].

We conducted Kruskal-Wallis tests to evaluate differences across clinical groups in cytokine abundance. To explore the relationship between cytokine abundance and other continuous variables (FEV1 and microbial alpha diversity), we employed Spearman's rank correlation coefficient after omitting outliers, defined as points lying more than 3 standard deviations away from the mean for each biomarker. Outlier removal was performed independently for each biomarker. Specifically, one outlier was removed for each of IL1RL1, IL17, CXCL9, WFDC2, DKK3, and CCL5. We applied linear mixed-effect models, adjusting for active cGVHD at enrollment, to evaluate the associations between each biomarker level and changes in FEV1%, FVC%, or DLCO over time, accounting for individual variability among subjects. Additionally, we used linear regression models, also adjusted for active cGVHD at enrollment, to examine the association between the change in cytokine abundance and the change in FEV1%, FVC%, or DLCO from baseline to 6 months. For examining the relationship between each cytokine and the microbial composition's beta diversity, we utilized MiRKAT [18] for baseline analysis. Since we considered a focused panel of biomarkers with known clinical relevance, we did not adjust for multiplicity across them. Due to the high dimensionality of the microbiome, we focused on kernel association methods to avoid the burden of multiple testing. A *P* value of $< .05$ was considered statistically significant.

RESULTS

Participant Characteristics

A total of 31 subjects with lung injury were enrolled, as were 12 controls. A total of 55.8% of the subjects were female. The average age of the total cohort was 49 years of age at the time of enrollment. Subjects had a median follow-up of 5.5 months. Subjects primarily self-identified as White (65%, n = 28), followed by LatinX (18.6%, n = 8), Asian (13.9%, n = 6), and Black (2.3%, n = 1) (Table 1). As expected, subjects in the control group had significantly higher initial FEV1% at enrollment when compared with subjects in the pre-BOS, transient pulmonary impairment, and BOS groups (Kruskal-Wallis $P < .001$). Pre-BOS and transient pulmonary impairment patients did not differ from each other in initial FEV1%, and both had higher initial FEV1% when compared with BOS subjects (Wilcoxon $P = .0017$, $P = .0013$, respectively). No patients in the control and transient impairment groups developed BOS during the duration of this study.

Lung Impairment and Serum Biomarkers

At baseline, of the 9 studied biomarkers, only CCL5 differed between groups ($P = .044$), with significantly lower levels in the BOS group ($P = .007$) when compared with the transient pulmonary impairment group. However, there was no difference in CCl5 levels between BOS, pre-BOS, or transient pulmonary impairment when compared with controls ($P = .90$, .63, and .17, respectively).

In analyses of baseline values with all groups combined, IL1RL1 and CXCL9 levels were negatively and significantly correlated with baseline FEV1% (IL1RL1 R = −0.4, $P = .013$; CXCL9 R = −0.33, $P = .044$; Figure 1). All other measured cytokines were not associated with baseline FEV1%. When only evaluating the BOS and pre-BOS groups combined, there were no associations between cytokines and FEV1% (Supplementary Figure 1). When evaluating associations with FVC% and all groups combined, IL1RL1 and DKK3 levels were negatively and significantly correlated with baseline FVC% (IL1RL1 R = −0.39, $P = .012$; DKK3 R = −0.45, $P = .004$; Supplementary Figure 2). When BOS and pre-BOS groups were combined, only DKK3 was negatively associated with FVC% (R = −0.49, $P = .02$; Supplementary Figure 3). With all groups combined, baseline DLCO was negatively correlated with CXCL9, WFDC2, and DKK3 (CXCL9 R = −0.34, $P = .041$; WFDC2 R = −0.34, $P = .035$; DKK3 R = −0.36, $P = .027$; Supplementary Figure 4), but only CXCL9 remained significantly associated with DLCO when only analyzing BOS and pre-BOS groups (Supplementary Figure 5).

We further constructed linear mixed-effects models to evaluate the associations between biomarker levels and change in FEV1%, FVC%, or DLCO over time when adjusted for active cGVHD at enrollment. IL1RL1 (coefficient = −0.288, $P = .004$) and IL-17 (coefficient = −4.44, $P = .03$) were negatively and significantly correlated with FEV1% change over time, while CCL5 was positively correlated with FEV1% change over time (coefficient = 0.73, $P = .03$; Table 2). When adjusted for active cGVHD at time of enrollment, IL1RL1 (coefficient −0.2, $P = .006$) was negatively and significantly associated with change in FVC%. After adjustment for cGVHD, no biomarkers were associated with a decrease in DLCO over time.

Additional linear regression models, also adjusted for active cGVHD at enrollment, examined the 6-month changes for each biomarker and the changes in FEV1%, FVC%, and DLCO. The results revealed significant negative correlations between the changes in FEV1% among all groups combined and IL-17 (coefficient = −5.466, $P = .012$; Figure 2). In other words, greater increases in IL-17 were associated with greater declines in FEV1% from enrollment to 6 months. When limited to only the BOS and pre-BOS groups, this effect remained significant (IL-17 coefficient −7.16, $P = .005$) and CCL5 positively

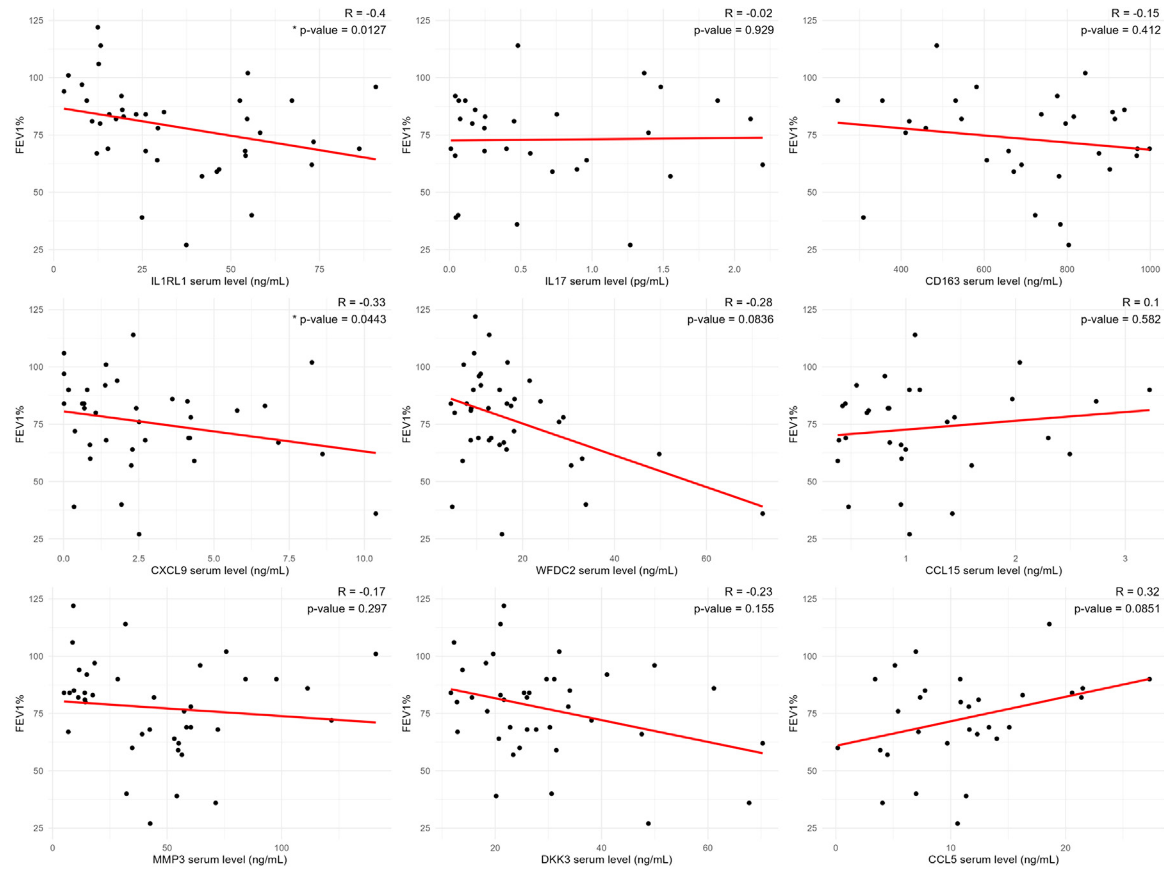

Figure 1. FEV1% correlates with inflammatory biomarkers in hematopoietic stem cell transplant recipients. * indicates $P < .05$ via the Spearman correlation coefficient.

correlated with changes in FEV1% (coefficient 2.27, P = .009; Supplementary Figure 6). FVC% showed a negative association with IL-17 (coefficient = −4.432, P = .008). There were no significant biomarker associations for the 6-month change in DLCO.

Associations of Stool Microbiome with Lung Impairment and Serum Biomarkers

Neither alpha diversity nor beta diversity of stool sample microbiomes differed among lung impairment groups. Stool alpha and beta diversities were not associated with a history of acute GVHD (aGVHD) or use of antibiotics preceding sample collection. Patients with active cGVHD at time of enrollment had higher stool alpha diversity (Simpson reciprocal P = .035) but no differences in beta diversity. At baseline, with all groups combined, we did not find a significant association between any serum biomarkers and alpha diversity (Supplementary Figure 7). This held true when only the BOS and pre-BOS groups were analyzed (Supplementary Figure 8). Beta diversity did not differ with biomarkers when all groups were combined (Supplementary Figure 9) or when only the BOS and pre-BOS groups were analyzed (Supplementary Figure 10).

Associations of Oral Microbiome with Lung Impairment and Serum Biomarkers

Alpha diversity of oral microbiome samples differed significantly between lung impairment groups, with BOS subjects demonstrating less alpha diversity than all other groups (Kruskal-Wallis P = .0057; Figure 3). Of note, and similar to analyses with stool samples, there was no difference in alpha or beta diversity between subjects who had or did not have aGVHD, nor was there an impact of antibiotic exposure. Oral beta diversity was not impacted by active cGVHD at time of enrollment (P = .095) and did not differ based upon FEV1% (P = .053) or clinical grouping at enrollment (Figure 3). With all groups combined, oral microbiome beta diversity differed significantly based upon levels of IL1RL1 (P = .038), WFDC2 (P = .036), CXCL9 (P = .009), DKK3 (P = .037), and CCL15 (P = .008; Figure 4). When only the BOS and pre-BOS groups were analyzed, oral microbiome beta diversity in all associations remained significant, except that beta diversity no

Table 2
Linear Mixed Effects Models Evaluating the Associations Between Biomarker Serum Levels and Change in FEV1% Over Time, Adjusted for Active cGVHD at Enrollment

Serum Biomarker	Coefficient Estimate	Standard Error	Akaike information criterion (AIC)	Bayesian Information Criterion (BIC)	Random Effect (Intercept)	Residual Standard Deviation	Degrees of Freedom	T value	*P* value
IL1RL1	−0.288	0.085	518.382	530.534	93.961	10.418	17	−3.378	.004*
IL17	−4.437	1.867	430.488	441.461	85.979	10.092	16	−2.376	.030*
CD163	−0.029	0.016	440.978	451.950	103.956	11.793	16	−1.840	.085
CXCL9	−0.228	0.867	513.145	525.189	84.110	11.935	17	−0.263	.795
WFDC2	−0.224	0.141	525.457	537.609	90.645	11.170	17	−1.582	.132
CCL15	1.528	4.924	432.719	443.691	78.591	12.241	16	0.310	.760
MMP3	−0.061	0.090	528.283	540.435	88.377	11.813	17	−0.679	.506
DKK3	−0.334	0.172	523.814	535.966	95.571	11.103	17	−1.943	.069
CCL5	0.729	0.311	433.048	444.020	68.203	11.354	16	2.344	.032*

* Indicates $P < .05$.

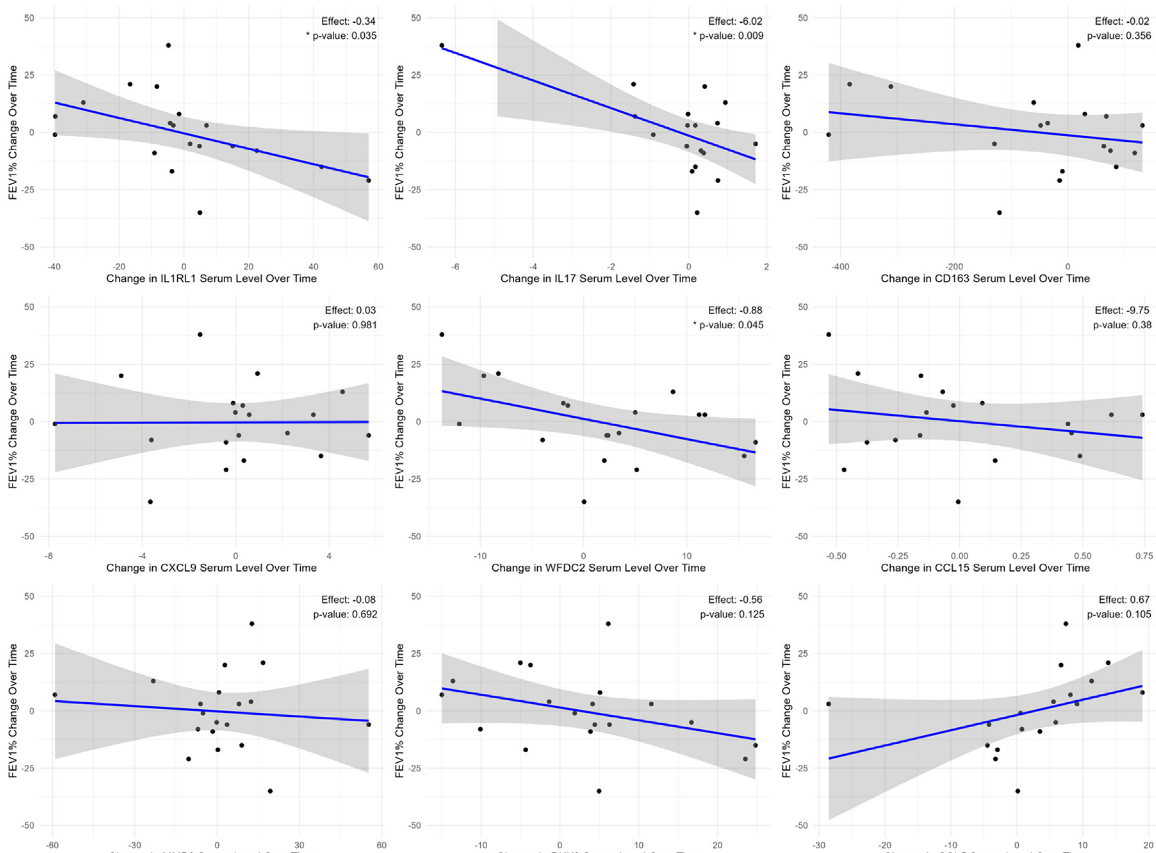

Figure 2. FEV1% varies over time, with variations in IL-17. * indicates $P < .05$.

longer was associated with CCL5 levels (Supplementary Figure 11).

DISCUSSION

In this study, we found that baseline IL1RL1 and CXCL9 levels were associated with progressive pulmonary impairment over time whereas CCL5 levels positively correlated with FEV1% and serum biomarkers did not differentiate BOS from patients with transient or no lung impairment. In addition, increases in IL-17 levels were associated with more pulmonary impairment over time. Furthermore, we show that pulmonary impairment is associated with greater disturbances in the oral

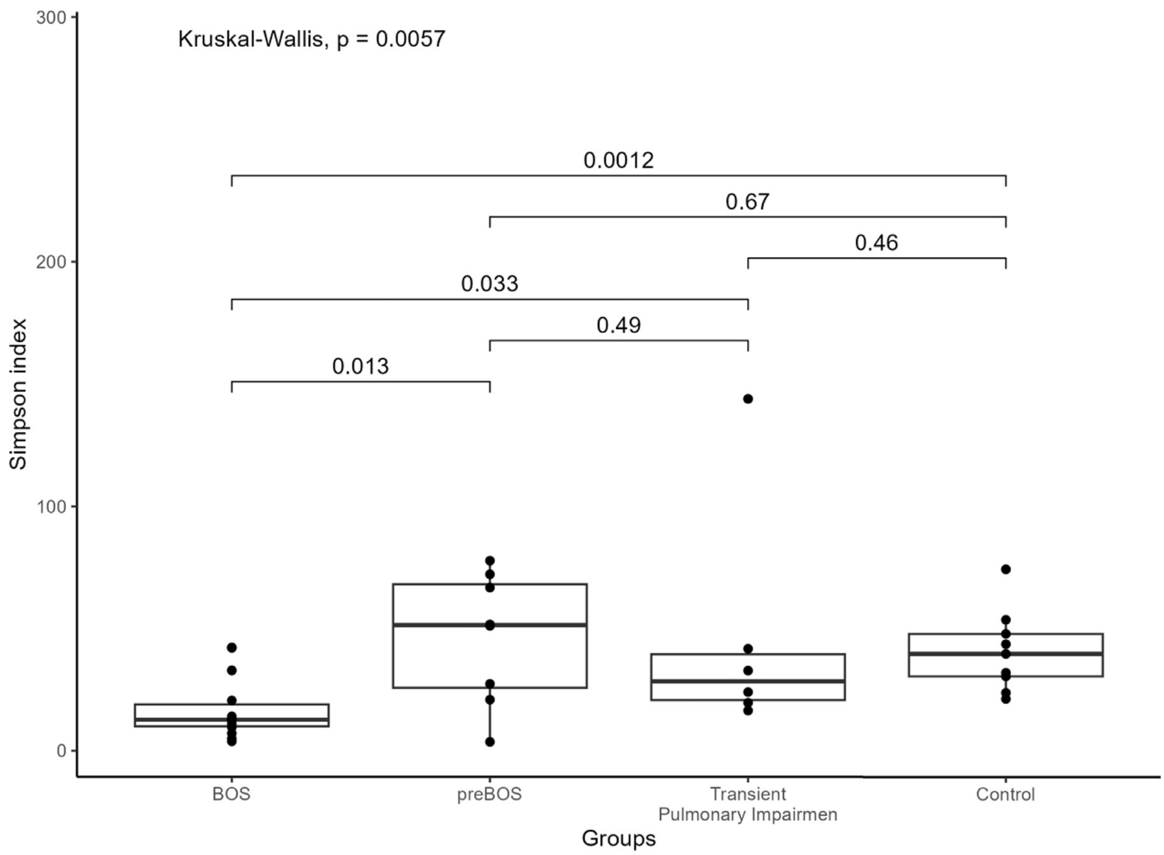

Figure 3. Oral microbiome alpha diversity is lower in hematopoetic stem cell recipients with bronchiolitis obliterans.

microbiome as compared with the stool microbiome. Participants with BOS had lower alpha diversity in the oral but not stool microbiome compared with other groups. While stool microbiome characteristics were only minimally associated with 1 baseline cytokine level, we found that baseline cytokine levels were associated with more dysbiosis in the oral microbiome. Our study suggests that the primary value of serum GVHD markers in patients with BOS is to identify patients at high risk for lung function decline over time. In addition, we show that oral microbiome changes may be sensitive to respiratory tract inflammation and serve as a marker of disease activity.

The morbidity and mortality associated with BOS may be potentially mitigated by early intervention [5,6], although no prospective study has shown this. As a result, National Institutes of Health (NIH) guidelines have advocated for aggressive surveillance and biomarkers to diagnose BOS, but to date those have been lacking [19]. This was a prospective, non-interventional pilot study in which we did not aim to uncover causative relationships between BOS and biomarkers. Rather, we hoped to discover biomarkers that may have such a relationship and that warrant further investigations into causality and pathogenesis—indeed, we did find several such biomarkers. In this study, we found that lower levels of CCL5 were associated with BOS when compared with the transient lung impairment group. When examining pulmonary impairment as a continuous variable, we found that higher baseline levels of IL1RL1 and CXCL9 were associated with more pulmonary impairment at baseline, higher levels of IL1RL1, DKK3, and IL-17 at baseline were associated with more pulmonary impairment over time, and increases in IL1RL1 and IL-17 were associated with greater declines in pulmonary function over time. Though these findings should be considered hypothesis generating given the exploratory nature of the study, each biomarker has the potential to be validated in prospective studies as markers of incident BOS or progressive BOS.

The current understanding of cGVHD is that the disease pathobiology is defined by a broad immune response resulting in inflammation, end-organ injury, and eventually fibrosis [15,19−22]. IL-33 is a potent regulator of T cell−mediated responses and can be considered an alarmin that can be released after tissue injury to a variety of organs, including the lung airways [23]. IL-33 is

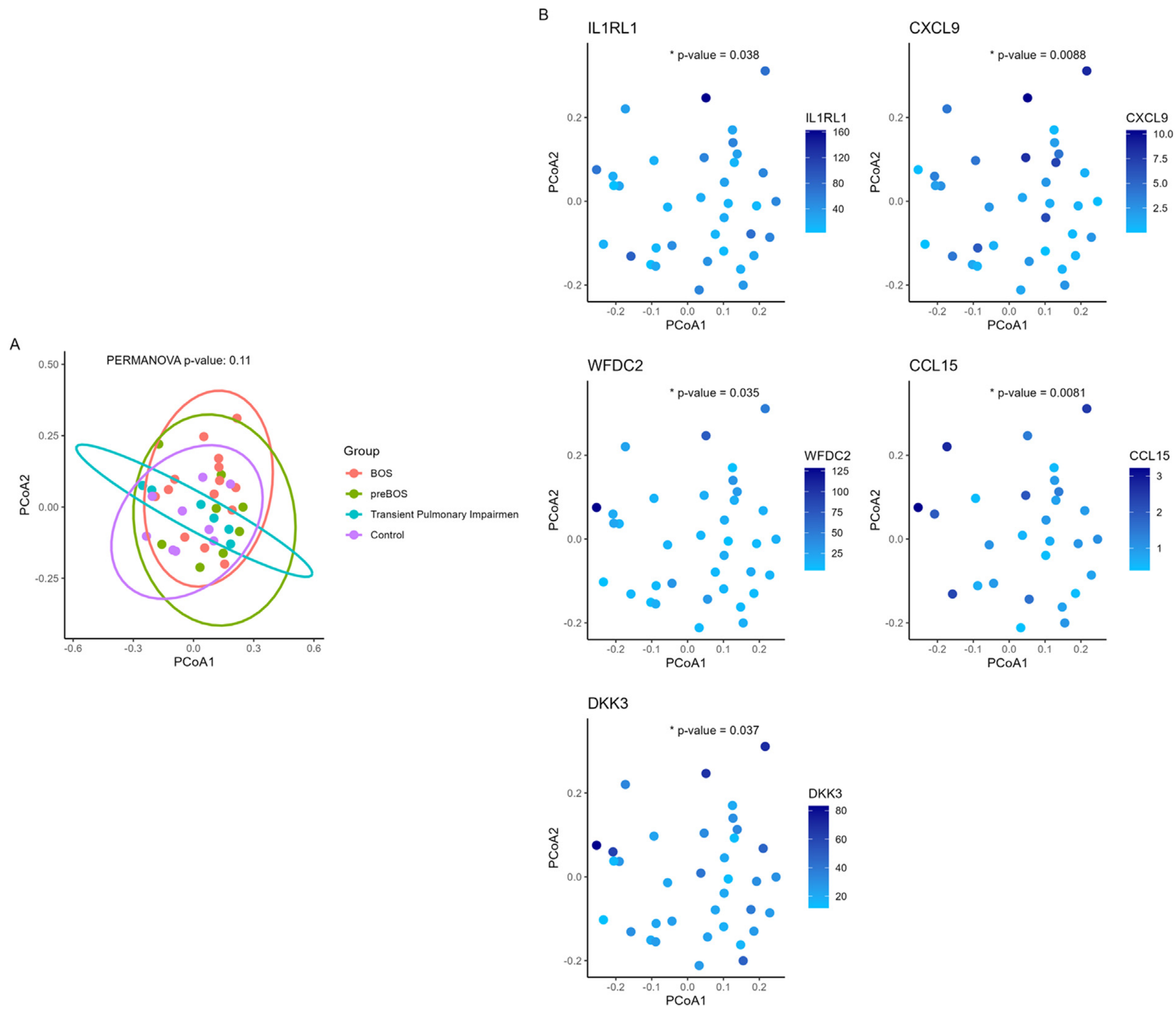

Figure 4. Oral microbiome beta diversity does not differ by lung injury group (A) but does correlate with several serum inflammatory biomarker levels (B). * indicates $P < .05$.

difficult to measure directly in serum owing to rapid turnover [24], but soluble IL1RL1, the decoy receptor for IL-33, is measurable in the serum and high circulating levels have been associated with cGVHD [22,25]. In this study, higher IL1RL1 levels were associated with more pulmonary impairment at baseline and progressive pulmonary impairment over time. Possibly due to the small sample size, we were not able to show that IL1RL1 was higher in BOS specifically; therefore, this will need to be validated prospectively. While the activation of the IL-33/IL1RL1 pathway has pleiotropic effects, it most notably induces a type 2 inflammatory/immune response and initiates a pro-fibrotic pulmonary cascade. These effects are consistent with the known pathobiology of BOS. We also found evidence of type 1 inflammation (CXCL9) being associated with pulmonary impairment at baseline, consistent with prior studies showing that elevated CXCL9 levels were associated with *de novo* cGVHD [26].

The association of IL-17 with pulmonary impairment at baseline and progressive pulmonary impairment is intriguing and is also supported by preclinical models. IL-17A has been implicated in the development of certain autoimmune diseases, such as psoriatic arthritis, but it also has protective roles against a broad range of infections and is involved in the repair of mucosal surfaces [27]. In a murine model of pulmonary cGVHD, the development of BOS was dependent on IL-17, and knockout of the retinoid-related orphan receptor γ gene, which is necessary for the development of Th17 cells, reduced the level of pulmonary impairment [28]. In that model, murine lung cGVHD was also dependent on CSF-1, which we did not evaluate in the current study. Both IL-17 and IL-33 are “druggable” targets and would be rational targets to study in patients with new-onset BOS in addition to ongoing trials of CSF-1 inhibition [29].

We also found higher levels of baseline WFDC2 (WAP Four-Disulfide Core Domain 2) with greater decreases in FVC and DLCO. WFDC2 is expressed in pulmonary epithelial cells [30−32], and *in vitro* it helps mediate the NF-kB pathway; overexpression of WFDC2 increases IL-6 expression. Elevated WFDC2 levels are seen in other lung conditions, including cystic fibrosis, interstitial lung disease, and chronic obstructive pulmonary disease, altogether suggesting a possible mechanism for its involvement in BOS-related inflammation and fibrosis [32−38]. The hemoglobin/haptoglobin scavenger receptor CD163 is expressed in macrophages and monocytes, in which it is thought to serve an anti-inflammatory role [39]; elevated levels have been associated with subsequent development of cGVHD [40]. DKK3 (dickkopf WNT signaling pathway inhibitor 3) inhibits WNT signaling in numerous organs, including the pulmonary epithelium, and is also a marker of cGVHD [41]. However, while other DKK proteins have been associated with pulmonary fibrosis, DKK3 was not. The effects of DKK3 are most pronounced in other organs, such as the kidney, heart, and skin [42−45]. Interestingly, we found that DKK3 was associated with beta diversity in the oral microbiome.

CCL5 is a chemokine known to attract CCR1 T cells and other immune cells to organ sites of cGVHD in murine models [46,47], but once T cells have infiltrated a target organ, counterregulatory pathways may induce a negative feedback loop that reduces CCL5 levels. Alternatively, because BOS generally occurs later than other cGVHD syndromes, it is possible that CCL5 is decreased due to GVHD therapies. However, that decrease is not sufficient to halt the incidence of BOS. Because of our study design, it is not possible to determine whether the observation of lower CCL5 in patients with BOS is a result of the small sample or due to GVHD treatment, or due to negative feedback pathways.

In general, we found that the effects of inflammatory cGVHD markers were seen in the oral microbiome but only minimally in the stool microbiome, indicating that systemic inflammation associated with pulmonary impairment may also influence dysbiosis in the oropharynx. The mechanism underpinning this observation is not entirely clear, nor is the direction of causation. Other studies have investigated the link between gut dysbiosis and cGVHD [48−50] but we found no significant associations between stool microbiota and pulmonary impairment, though the sample size is likely insufficient to capture smaller effects. On the other hand, in our study we found that those with newly diagnosed BOS had lower oral microbiome alpha diversity than all other groups. The pulmonary microbiome is influenced heavily by oral microbiome composition [51], but future studies will need to investigate whether a similar level of dysbiosis occurs in the pulmonary microbiome [52]. Interestingly, a loss of pre-HCT microbial diversity in the lung has been associated with an increased risk of fatal pulmonary complications after HCT in children. While we did not see differences in beta diversity among groups, we found that several inflammatory markers were associated with beta diversity, indicating a link between systemic inflammation and oral dysbiosis. These findings require prospective validation, but indicate that studies involving the microbiome in BOS should focus on the oral and respiratory tract.

The current study is limited by the small sample size and, more specifically, the small number of subjects in the control group. Our smaller sample size likely limited the differences in microbiome diversity that we were able to identify. Additionally, most of the markers in this study did not vary by group. Rather, they were associated with the degree of pulmonary impairment, which may have more statistical power to capture associations. Therefore, the findings must be considered exploratory and require validation in larger, prospective studies. One source of variation might be that patients with BOS may have had varying lengths of time prior to diagnosis, since PFTs were performed every few months as per NIH guidelines, a strategy that is likely to be insufficient to capture BOS immediately upon diagnosis. In addition, patients were enrolled in the study at varying time points post-transplant; therefore, our ability to determine at what point these changes first manifest is limited. It is possible that the changes in biomarkers noted in this study could be due to changes in severity of non-pulmonary cGVHD, which can be more challenging to quantify given the lack of a granular continuous measure like FEV_1. Ongoing prospective observational studies will be well suited to validating our findings and better delineate the timeline of clinical development of lung impairment as compared with biochemical detection of inflammatory markers.

In conclusion, we demonstrate the association of several inflammatory cGVHD markers with new and progressive pulmonary impairment, and show that certain inflammatory markers influence oral microbiome composition. In addition, we show that oral microbiome diversity is lower in patients with BOS. Our findings require validation but offer further insights into new potential therapies for BOS. Furthermore, with validation, these serum markers may offer an opportunity to diagnose new BOS against the background of cGVHD and offer patients a noninvasive approach to improving current methods of surveillance.

ACKNOWLEDGMENTS

ZW, RRJ, and CBP were funded by National Institutes of Health/National Heart, Lung, and Blood Institute grant no. R01 HL158796. AS was funded by National Institutes of Health/National Institute of Allergy and Infectious Diseases grant no. K23 AI117024. SP was funded by Medical University of South Carolina grant nos. R01 HL158096 and R01 CA264921. The Microbiome Facility was supported in part by The University of Texas MD Anderson Cancer Center and P30CA016672.

DATA SHARING STATEMENT

Data will be shared upon reasonable request by contacting the corresponding author.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jtct.2025.06.020.

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